Reinforcement Learning for Global Equity ETF Allocation

Roshni Guha

Agenda

1. Motivation
2. Blended Momentum Signal
3. Model Architecture
4. Training
5. Results
6. How It Works

Motivation

Problem: Global equity regions respond differently to macro conditions.

Regional ETFs (actions)

- SPY-US – U.S. equities
- EWJ-US – Japan
- INDA-US – India
- MCHI-US – China
- EZU-US – Eurozone
- BIL-US – Cash

Macro signals (state)

- TLT – Rates / risk-off
- GLD – Inflation / safe haven
- USO – Oil / global growth
- SLV – Industrial demand
- VIXY – Volatility / fear
- IBIT – Risk appetite / crypto

Goal: Learn a policy that rotates between regions based on macro regime, outperforming static equal-weight allocation.

Blended Momentum Signal (bmom)

Rather than using 3 raw momentum windows as separate features, we construct a **single blended signal** per asset:

$$\mathtt{bmom}_t^{(i)} = 0.15 \cdot m_{10}^{(i)} + 0.35 \cdot m_{20}^{(i)} + 0.35 \cdot m_{40}^{(i)} + 0.15 \cdot m_{60}^{(i)}$$

Window	Weight	Role
10 days	15%	Recent signal (noisy)
20 days	35%	Short-medium trend
40 days	35%	Medium trend
60 days	15%	Longer-term (stale)

Benefits:

- State dimension 18 $\rightarrow$ 6 (fewer parameters, less overfitting)
- Center windows dominate $\Rightarrow$ smoother, less whipsaw rotation
- Single interpretable feature per macro asset

State & Action Space

State vector $s_t \in \mathbb{R}^6$:

$$s_t = \left[\, \mathtt{bmom}^{(\mathrm{TLT})},\ \mathtt{bmom}^{(\mathrm{GLD})},\ \mathtt{bmom}^{(\mathrm{USO})},\ \mathtt{bmom}^{(\mathrm{SLV})},\ \mathtt{bmom}^{(\mathrm{VIXY})},\ \mathtt{bmom}^{(\mathrm{IBIT})} \,\right]^\top$$

Action space (discrete, 6 options):

$$a_t \ \in\ \{\text{SPY-US},\ \text{EWJ-US},\ \text{INDA-US},\ \text{MCHI-US},\ \text{EZU-US},\ \text{BIL-US}\}$$

The agent allocates **100%** to one ETF each day.

Reward function:

$$r_t \ = \ R_t \ - \ \lambda\, \sigma_t, \qquad \lambda = 0.5$$

- R_t: next-day return of chosen ETF
- σ_t: rolling 20-day volatility of chosen ETF

Policy Network Architecture

Policy network $\pi_\theta(a \mid s)$:

$$s \xrightarrow{\text{FC-128, ReLU}} \xrightarrow{\text{FC-128, ReLU}} \xrightarrow{\text{FC-6, Softmax}} P(a)$$

Value network $V_\phi(s)$:

$$s \xrightarrow{\text{FC-128, ReLU}} \xrightarrow{\text{FC-128, ReLU}} \xrightarrow{\text{FC-1}} \hat{v}$$

Objective:

$$\pi^* = \arg\max_\pi \mathbb{E}_\pi \left[\sum_t \gamma^t r_t \right], \quad \gamma = 0.95$$

Hyperparameters

Parameter	Value
Policy LR	3×10^{-4}
Value LR	1×10^{-3}
Hidden dim	128
Clip ε	0.2
GAE λ	0.95
Entropy coef	0.01
Grad clip	0.5

Training: Proximal Policy Optimization (PPO)

PPO update loop (50 epochs):

1. **Collect trajectories**: simulate full dataset, record (s_t, a_t, r_t)
2. **Compute GAE advantages**:

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}, \qquad \delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$$

3. **Update policy** with clipped objective:

$$\mathcal{L}^{\text{PPO}} = \mathbb{E}_t \left[\min\left(\rho_t \hat{A}_t,\ \text{clip}(\rho_t, 0.8, 1.2)\, \hat{A}_t \right) \right], \quad \rho_t = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$$

4. **Update value network** with MSE loss

Why PPO? Clipping prevents large, destabilising policy updates while still making efficient use of collected experience.

Validation Methodology

Period	Dates	Days
Training	Jan 2004 – Dec 2022	≈5,457
Test (OOS)	Jan 2022 – Sep 2025	≈3 years

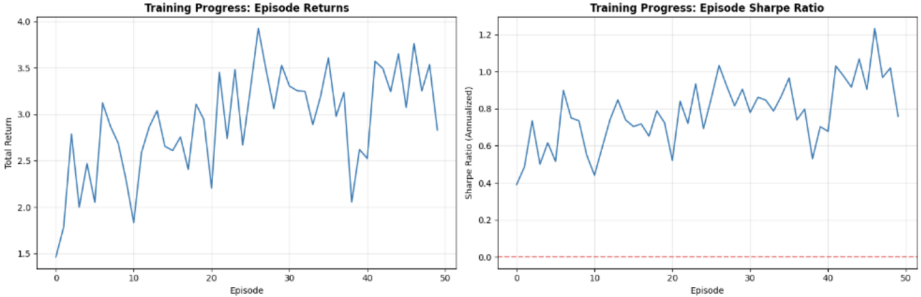
Baseline: Equal-weight portfolio — 16.67% to each of the 6 trade ETFs, rebalanced daily.

Evaluation metrics:

- Cumulative return and equity curve
- Risk-adjusted return (Sharpe-like)
- Drawdown profile
- Allocation frequency by ETF

Training Metrics

```
Training on 5457 days of data...
State dimension: 18, Action dimension: 6
Training for 50 episodes...
Pre-computing returns matrix...
Sample returns for SPY-US: [0.010878304078312695, 0.000978655818617557 6, 0.003376290508406265 5]
Pre-computing state features...
Starting training loop...
Episode 10/50 | Avg Return (last 10): 2.3535 | Avg Sharpe (last 10): 0.6175
Episode 20/50 | Avg Return (last 10): 2.6804 | Avg Sharpe (last 10): 0.6946
Episode 30/50 | Avg Return (last 10): 3.1822 | Avg Sharpe (last 10): 0.8236
Episode 40/50 | Avg Return (last 10): 3.8381 | Avg Sharpe (last 10): 0.7876
Episode 50/50 | Avg Return (last 10): 3.2935 | Avg Sharpe (last 10): 0.9550
```




```
Training completed!
Final Sharpe Ratio: 0.7589

Training completed!
```

Figure: Average reward, policy loss, and value loss over 50 training epochs.

Backtest: RL vs. Equal-Weight Baseline

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METRICS COMPARISON
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RL_GlobalEquity:
----------------------------------------
  Total Return: 22.445012
  Annualized Return: 0.156826
  Annualized Volatility: 0.174302
  Sharpe Ratio: 0.899736

Baseline_EqualWeight:
----------------------------------------
  Total Return: 2.237195
  Annualized Return: 0.055745
  Annualized Volatility: 0.135772
  Sharpe Ratio: 0.410582

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EQUITY CURVES
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RL_GlobalEquity:
  Final Value: $23.45
  Total Return: 2244.50%
```

Equity Curves

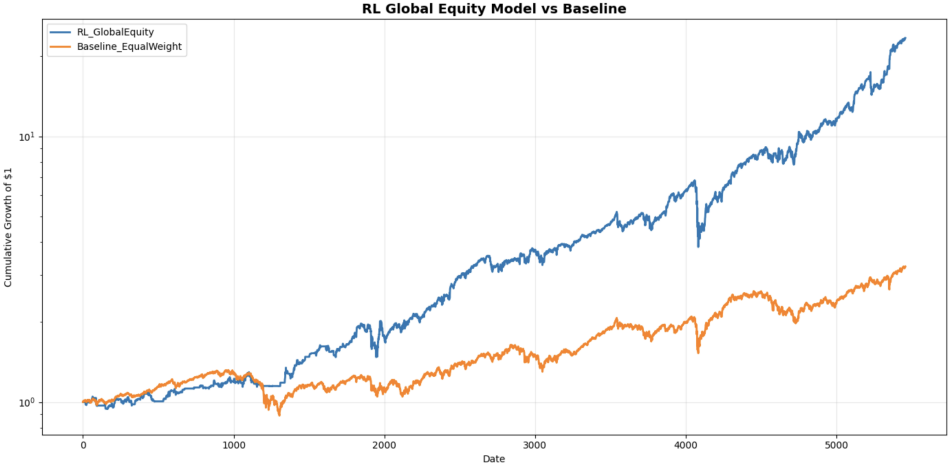

Figure: Cumulative growth of $1 — RL model vs. equal-weight baseline.

Key Findings

- **Improved risk-adjusted returns** vs. static equal-weight allocation
- **Regime-adaptive** — rotates between regions and cash based on macro `bmom`
- **Out-of-sample generalisation** on 2022–2025 test set
- **Blended momentum** reduces state dim from 18 $\rightarrow$ 6 without sacrificing signal quality
- **Learned rules** (examples):
  - Positive commodity `bmom` $\Rightarrow$ EWJ-US, EZU-US
  - Positive VIXY `bmom` $\Rightarrow$ BIL-US (cash)
  - Stable macro signals $\Rightarrow$ SPY-US

Summary

What we built

- PPO-based macro rotation model
- Blended momentum state (`bmom`)
- 6-asset discrete action space
- Risk-penalised reward ($\lambda = 0.5$)

What we found

- RL learns meaningful macro regimes
- `bmom` weights improve stability
- Outperforms equal-weight baseline
- Generalises out-of-sample

Questions?